Jan2024 MTH208 EQP Final Solution (For Student's Reference)

1a) The Jordan canonical form of is

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}.$$

1b) Hence the number of possible Jordan canonical forms is $1 \times 1 \times 4 \times 3 = 12$.

2c(ii)

$$\langle\langle x, y\rangle\rangle_A = 0$$

$$y^T x = -3.$$

2c(iii)

The required orthogonal basis is
$\{(1, -3, -1)^T, (-2, 0, 1)^T, (120/17, -15, -60/17)^T\}$.

3)

The rank of f is $2 + 2 = 4$, and its signature is $2 - 2 = 0$

4a)

$$Q = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 39 & 0 \\ 0 & 34 \end{pmatrix}$$

4b)

$$\Sigma = \begin{pmatrix} \sqrt{39} & 0 \\ 0 & \sqrt{34} \\ 0 & 0 \end{pmatrix}$$

$$V = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

$$U = \frac{1}{\sqrt{6630}} \begin{pmatrix} 13\sqrt{34} & -\sqrt{39} & 13\sqrt{5} \\ \sqrt{34} & -12\sqrt{39} & -14\sqrt{5} \\ 5\sqrt{34} & 5\sqrt{39} & -31\sqrt{5} \end{pmatrix}$$

$A = U\Sigma V^*$ is a singular value decomposition of A

5a)

$$W_1 = \text{span}\{(2, -2, 1)^T\}.$$

$$W_2 = \text{span}\{(1, 0, 0)^T, (0, -1, 1)^T\}.$$

5b)

$$(I - A)^2 = \begin{pmatrix} (-3)^n & 2(-3)^n - 2 & 2(-3)^n - 2 \\ 0 & -(-3)^n + 2 & -2(-3)^n + 2 \\ 0 & (-3)^n - 1 & 2(-3)^n - 1 \end{pmatrix}.$$

5c)

A and B will be similar over $\mathbb{R}$.

6a(i)

The dual basis $\mathcal{B}^* = \{f_1, f_2\}$,

$$f_1((x_1, x_2, x_3, x_4)^T) = (-1/13)x_1 + (-5/13)x_2$$

$$f_2((x_1, x_2, x_3, x_4)^T) = (-3/13)x_1 + (-2/13)x_2.$$

6a(ii)

A basis $\{g_1, g_2\}$ of W^0,

$$g_1((x_1, x_2, x_3, x_4)^T) = (-1/13)x_1 + (21/13)x_2 + x_3$$

$$g_2((x_1, x_2, x_3, x_4)^T) = (31/13)x_1 + (38/13)x_2 + x_4.$$

6b(ii)

$$\begin{pmatrix} -122/47 & 115/47 \\ 135/47 & -254/47 \end{pmatrix}.$$